

## Prerequisites for Canonical Form.

Def. Let  $T: V \rightarrow V$  be a linear operator on  $V$ .

A subspace  $W \subseteq V$  is called  $T$ -invariant if:  $T[W] \subseteq W$ .

Note:  $\text{Im } T$ ,  $\ker T$ ,  $E_\lambda$  are all  $T$ -invariant.

Lemma. A  $T$ -cyclic subspace generated by  $x \in V$

$$W = \langle T^n(x) \mid n = 0, 1, 2, \dots \rangle$$

is the smallest  $T$ -invariant containing  $x \in V$ .

Proof.  $T$ -invariant is clear. Suppose that  $W'$  is a  $T$ -invariant subspace containing  $x \in V$ .

Then,  $x \in W' \Rightarrow T(x) \in W' \Rightarrow T^2(x) \in W' \dots$

By induction, for all  $n \in \mathbb{N}$ ,  $T^n(x) \in W'$ . That is,  $W'$  contains generating set of  $W$ .

Lemma. If  $W$  is  $T$ -invariant, then also  $(T - cI)$ -invariant, for any  $c \in F$ .

Proof. Let  $x \in W$  be given. Then,

$$(T - cI)(x) = \underbrace{T(x)}_{\in W} - c \underbrace{x}_{\in W} \in W$$

## Canonical Form.

Def. Let  $T: V \rightarrow V$  be a linear operator, and  $\lambda \in F$  be a scalar.

A non-zero vector  $x \in V$  is called Generalized eigenvector of  $T$  corresponding to  $\lambda$  if;

$$\text{for some } p \in \mathbb{N}, (T - \lambda I)^p(x) = 0$$

Denote a set of generalized eigenvectors of  $\lambda$  as  $K_\lambda$ . ( $K_\lambda = \bigcup_{n=1}^{\infty} \ker(T - \lambda I)^n$ )

Note: If a scalar  $\lambda \in F$  has a generalized eigenvector s.t.  $(T - \lambda I)^p(0) = 0$ , then  $\lambda$  is eigenvector of  $T$ , corresponding the eigenvector  $(T - \lambda I)^{p-1}(x)$ .

Thm. Let  $T: V \rightarrow V$  be a linear operator, and  $\lambda \in F$  be an eigenvalue of  $T$ .

1) Generalized Eigenspace  $K_\lambda$  is  $T$ -invariant subspace which contains  $E_\lambda$ .

2) For any eigenvalue  $\mu \in F$  with  $\mu \neq \lambda$ ,  $K_\mu \cap K_\lambda = \{0\}$ .

3) For any scalar  $\gamma \in F$  with  $\gamma \neq \lambda$ , the restriction

$$(T - \lambda I)|_{K_\lambda}: K_\lambda \rightarrow K_\lambda; x \mapsto (T - \lambda I)(x)$$

is bijection (That is, Isomorphism)

Proof: Clearly,  $0 \in K_\lambda$ . To use subspace criterion, let  $c, \gamma \in K_\lambda$  be given.

That is, for some  $p, q \in \mathbb{N}$ ,  $(T - \lambda I)^p(x) = (T - \lambda I)^q(y) = 0$ . Now,

$$\begin{aligned} (T - \lambda I)^{p+q}(cx + \gamma) &= c \cdot (T - \lambda I)^{p+q}(x) + (T - \lambda I)^{p+q}(\gamma) \\ &= c \cdot (T - \lambda I)^q(0) + (T - \lambda I)^p(0) \end{aligned}$$

This implies  $cx + \gamma \in K_\lambda$ .

To show that  $T$ -invariant, let  $x \in K_\lambda$  s.t.  $(T - \lambda I)^p(x) = 0$ . Then

$$(T - \lambda I)^p(T(x)) = T \circ (T - \lambda I)^p(x) = T(0) = 0.$$

This implies  $T[K_\lambda] \subseteq K_\lambda$

Let  $\mu \in F$  be an eigenvalue of  $T$  s.t.  $\mu \neq \lambda$ . Choose  $x \in K_\mu \cap K_\lambda$ . Then, for some  $p, q \in \mathbb{N}$ ,

$$(T - \lambda I)^p(x) = (T - \mu I)^q(x) = 0.$$

In other words, for  $f_\lambda(t) = (t - \lambda)^p$ ,  $f_\mu(t) = (t - \mu)^q$ ,  $f_\lambda(T)(x) = f_\mu(T)(x) = 0$ .

Since  $F[t]$  is PID, it is also Bezout domain, there exist  $g_1(t), g_2(t) \in F[t]$  s.t.

$$g_1(t)f_\lambda(t) + g_2(t)f_\mu(t) = \gcd(f_\lambda(t), f_\mu(t)) = 1.$$

Therefore,  $g_1(T)f_\lambda(T)(x) + g_2(T)f_\mu(T)(x) = g_1(T)(0) + g_2(T)(0) = 0 = x$ .

Let  $\gamma \in F$  with  $\gamma \neq 1$  be given. Since  $K_\lambda$  is  $T$ -invariant, for any  $x \in K_\lambda$ ,

$$(T - \gamma I)(x) = T(x) - \gamma \cdot x \in K_\lambda$$

That is,  $K_\lambda$  is  $(T - \gamma I)$ -invariant.

Let  $x \in K_\lambda$ , and suppose  $(T - \gamma I)(x) = 0$ . Then,  $T(x) = \gamma x$ , thus  $x \in E_\gamma \subseteq K_\gamma$

But by 2.,  $K_\gamma \cap K_\lambda = \{0\}$ ,  $x = 0$ . That is,  $\ker(T - \gamma I)|_{K_\lambda} = \{0\}$ , thus  $1-1$ .

Let  $x \in K_\lambda$  be given. put  $p \in \mathbb{N}$  be the smallest Natural number s.t.  $(T - \lambda I)^p(x) = 0$ .

Define  $W = \langle \{x, (T - \lambda I)(x), \dots, (T - \lambda I)^{p-1}(x)\} \rangle$

Then,  $W$  equals to  $(T - \lambda I)$ -cyclic subspace generated by  $x$ ,

thus  $W$  is the smallest  $(T - \lambda I)$ -invariant subspace containing  $x$ , thus  $W \subseteq L_\lambda$ .

Also,  $W$  is  $(T - \gamma I)$ -invariant. Now, a restriction

$$(T - \gamma I)|_W : W \rightarrow W : \gamma \mapsto (T - \gamma I)(\gamma)$$

is well-defined injection, and surjection, since  $W$  is fin. dim.

Finally, there exists  $\gamma \in W$  s.t.  $(T - \gamma I)|_W(\gamma) = x$ , thus  $(T - \gamma I)|_{K_\lambda}$  is surjection

Thm. Let  $T: V \rightarrow V$  be a linear operator on fin. dim  $V$  over  $F$ .

If a characteristic polynomial  $\det(T - tI)$  splits over  $F$ ,

then for an eigenvalue  $\lambda$  with multiplicity  $m \geq 1$ , the followings are hold:

1)  $\dim K_\lambda \leq m$

2)  $K_\lambda = \ker(T - \lambda I)^m$

Proof. Denote  $f(t) = \det(T - tI)$  and  $h(t) = \det(T|_{K_\lambda} - tI)$ .

If  $T|_{K_\lambda}$  has another eigenvalue  $\mu$ , then there exists a non-zero vector  $v$  s.t.

$$v \in E_\mu \subseteq K_\mu \text{ and } v \in K_\lambda.$$

But, since  $K_\mu \cap K_\lambda = \{0\}$  it is contradiction. Now,

$$h(t) = (-1)^d (t - \lambda)^d$$

where  $d = \dim K_\lambda$ . Since  $h(t)$  divides  $f(t)$ ,  $\dim K_\lambda = d \leq m$ .

Since  $K_\lambda = \bigcup_{n=1}^{\infty} \ker(T - \lambda I)^n$ ,  $\ker(T - \lambda I)^m \subseteq K_\lambda$  is trivial.

And,  $f(t) = (t - \lambda)^m g(t)$  where  $g(t)$  has no factor  $(t - \lambda)$ .

For any  $\gamma \neq \lambda$ ,  $(T - \gamma I)|_{K_\lambda}$  is bijection. Thus,  $g(T)|_{K_\lambda}$  is bijection.

Let  $x \in K_\lambda$ . Then, for some  $y \in K_\lambda$ ,  $x = g(T)(y)$ . Now,

$$(T - \lambda I)^m(x) = (T - \lambda I)^m(g(T)(y)) = f(T)(y) = 0.$$

by Cayley-Hamilton Theorem. Thus  $x \in \ker(T - \lambda I)^m$ .

Thm. Let  $V$  be a fin. dim vector space,  $T: V \rightarrow V$  be a linear operator.

If a characteristic polynomial  $f(t)$  of  $T$  splits over  $F$ ,  
then for distinct eigenvalues  $\lambda_1, \dots, \lambda_k$  of  $T$ ,

$$V = K_{\lambda_1} \oplus \dots \oplus K_{\lambda_k}.$$

Proof. Denote  $n = \dim V$ , and

$$f(t) = (-1)^n (t - \lambda_1)^{n_1} \dots (t - \lambda_k)^{n_k}.$$

And for each  $1 \leq j \leq k$ , define

$$f_j(t) = \prod_{\substack{i \neq j \\ 1 \leq i \leq k}} (t - \lambda_i)^{n_i}$$

Since  $f_j$  ( $1 \leq j \leq k$ ) are mutually disjoint, there exist  $g_j(t)$  ( $1 \leq j \leq k$ ) such that

$$g_1(t)f_1(t) + \dots + g_k(t)f_k(t) = 1.$$

Now, for any  $x \in V$ ,

$$g_1(T)f_1(T)(x) + \dots + g_k(T)f_k(T)(x) = x.$$

Meanwhile for each  $g_j(T)f_j(T)(x)$ , the Cayley-Hamilton thm gives

$$\begin{aligned} & (T - \lambda_j I)^{n_j} (g_j(T)f_j(T)(x)) \\ &= (T - \lambda_j I)^{n_j} (f_j(T)g_j(T)(x)) \\ &= f(T)(g_j(T)(x)) \\ &= 0 \end{aligned}$$

That is,  $g_j(T)f_j(T)(x) \in K_{\lambda_j}$ . This proves  $V = K_{\lambda_1} + \dots + K_{\lambda_k}$ .

To show the uniqueness, note that since  $K_{\lambda_i}$  is  $T$ -invariant,  $f_j(T)$ -invariant.

Also, a restriction  $f_j(T)|_{K_{\lambda_j}}$  is bijection, being composition of bijections.

$$\text{Moreover, for } i \neq j, \quad f_j(T)[K_{\lambda_i}] = f_j(T)[\ker(T - \lambda_i I)^{n_i}] = \{0\}$$

Now, if  $\sum_{i=1}^k v_i = \sum_{i=1}^k w_i$  where  $v_i, w_i \in K_{\lambda_i}$  ( $1 \leq i \leq k$ ), then for each  $1 \leq j \leq k$ ,

$$f_j(T)(v_j) = \sum_{i=1}^k f_j(T)(v_i) = \sum_{i=1}^k f_j(T)(w_i) = f_j(T)(w_j).$$

Since  $f_j|_{K_{\lambda_j}}$  is injection,  $v_j = w_j$ .

Thm. Let  $V$  be a finite dimensional vector space,  $T: V \rightarrow V$  be a linear operator.  
If  $\det(T-tI)$  splits over  $F$  as

$$\det(T-tI) = (\lambda_1 - t)^{m_1} (\lambda_2 - t)^{m_2} \cdots (\lambda_k - t)^{m_k}.$$

Then,  $V = K_{\lambda_1} \oplus \cdots \oplus K_{\lambda_k}$ . Let  $\beta_{\lambda_i}$  be a basis for  $K_{\lambda_i}$ .

The following statements are true:

1)  $i \neq j \Rightarrow \beta_i \cap \beta_j = \emptyset$ .

2)  $\beta = \bigcup_{i=1}^k \beta_i$  is basis for  $V$

3) for each  $1 \leq i \leq k$ ,  $\dim K_{\lambda_i} = m_i$ .

Proof. 1) and 2) are clear. To show 3), see that:

$$\dim V = \sum_{i=1}^k m_i = \sum_{i=1}^k \dim(K_{\lambda_i})$$

And for all  $1 \leq i \leq k$ ,  $\dim K_{\lambda_i} \leq m_i$ , thus  $m_i = \dim K_{\lambda_i}$ .

Def. Let  $T: V \rightarrow V$  be a linear operator, and  $x \in V$  be a generalized eigenvector of  $T$  corresponding to  $\lambda \in F$ . For the smallest  $p \in \mathbb{N}$  s.t.  $(T - \lambda I)^p(x) = 0$ , a set

$$\{(T - \lambda I)^{p-1}(x), (T - \lambda I)^{p-2}(x), \dots, (T - \lambda I)(x), x\}$$

is called cycle of  $x$ , and denote  $C_x$ . (Note: only  $(T - \lambda I)^{p-1}(x)$  is eigenvector of  $T - \lambda I$ ).

Thm. Let  $T: V \rightarrow V$  be a linear operator,  $x \in V$  be a general. eigen. of  $T$  corresponding to  $\lambda \in F$ .

For cycle  $C_x$  of  $x$ ,  $\langle C_x \rangle$  is  $T$ -invariant, and  $[T|_{\langle C_x \rangle}]_{C_x}$  is Jordan block.

Proof. Denote  $C_x = \{(T - \lambda I)^{p-1}(x), \dots, x\}$ .

For each  $0 \leq i \leq p-1$ ,

$$(T - \lambda I)((T - \lambda I)^i(x)) = (T - \lambda I)^{i+1}(x) \in \langle C_x \rangle$$

Thus,  $\langle C_x \rangle$  is  $(T - \lambda I)$ -invariant, so that  $T$ -invariant.

Meanwhile,  $[T|_{\langle C_x \rangle}]_{C_x} = [T - \lambda I|_{\langle C_x \rangle}]_{C_x} + [\lambda I|_{\langle C_x \rangle}]_{C_x}$  and

$$[T - \lambda I|_{\langle C_x \rangle}]_{C_x} = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & 0 \end{bmatrix}$$

Gives the result.

Note:  $C_x = \{(T - \lambda I)^{p-1}(x), \dots, x\}$  are distinct, because

'If  $0 \leq i < j < p-1$  satisfies

$$(T - \lambda I)^i = (T - \lambda I)^j$$

$$\Rightarrow (T - \lambda I)^{\underbrace{p-j-1+i}_{p-1-(j-i)}} = (T - \lambda I)^{p-1} = 0$$

Contradiction.

Thm. Let  $\lambda \in F$  be an eigenvalue of  $T: V \rightarrow V$ .

Let  $x_1, \dots, x_b \in K_\lambda$  be distinct vectors.

If initial vector of each  $C_{x_i}$  are mutually different and initial vectors are linearly independent

then  $C_{x_i}$  are mutually disjoint and  $C = \bigcup_{i=1}^b C_{x_i}$  is linearly independent.

Proof. Denote that

$$C_x = \{(T - \lambda I)^{a-i}(x), \dots, x\}, \quad C_y = \{(T - \lambda I)^{b-j}(y), \dots, y\} \quad a \leq b$$

If for some  $1 \leq i \leq a$   $1 \leq j \leq b$

$$(T - \lambda I)^{a-i}(x) = (T - \lambda I)^{b-j}(y),$$

$$\text{Then } (T - \lambda I)^a(x) = (T - \lambda I)^{b-j+i}(y) = 0 = (T - \lambda I)^{a-i+j}(x)$$

If  $i-j=0$ , then these have same initial vector, contradiction.

If  $|i-j| > 0$ , then contradiction to minimality.

$$\begin{pmatrix} a & & \\ & b & \\ & & c \end{pmatrix}$$

$$\begin{pmatrix} a & 1 & & \\ & b & 1 & 0 \\ & & c & 1 \\ 0 & & & \ddots \end{pmatrix}$$

Let  $T: V \rightarrow V$  be a linear operator, and  $f(T)$  be a characteristic polynomial of  $T$ .

Def. Suppose there  $\lambda \in F$  be an eigenvalue of  $T$ . Define a set:

$$K_\lambda := \left\{ x \in V \mid (T - \lambda I)^p(x) = 0 \text{ for some } p \in \mathbb{N} \right\} = \bigcup_{n=1}^{\infty} \text{Ker}(T - \lambda I)^n$$

It is called a generalized eigenspace corresponding to  $\lambda$ .

For non-zero  $x \in K_\lambda$ , the smallest positive integer  $p \in \mathbb{Z}$  s.t.  $(T - \lambda I)^p(x) = 0$  is called period of  $x$ .  
(If  $x=0$ , period of  $x$  is assigned as zero.) (or length)

Lemma. Let  $\lambda \in F$  be an eigenvalue of  $T$ . Then, the following properties are satisfied:

- 1).  $K_\lambda$  is  $T$ -invariant subspace of  $V$  containing  $E_\lambda$ .
- 2). If  $\mu \in F$  is an eigenvalue of  $T$  s.t.  $\mu \neq \lambda$ , then  $K_\mu \cap K_\lambda = \{0\}$
- 3). For any CGF s.t.  $c \in F$ ,  $(T - cI)|_{K_\lambda}$  is a bijection.

Proof. Let  $x, y \in K_\lambda$  and  $\lambda \in F$  be given. Denote  $p, q \in \mathbb{Z}$  as period of  $x, y$ , respectively.

Then,

$$(T - \lambda I)^{p+q}(\alpha x + y) = \alpha (T - \lambda I)^{p+q}(x) + (T - \lambda I)^{p+q}(y) = 0.$$

Clearly,  $0 \in K_\lambda$ . Therefore, subspace criterion gives  $K_\lambda \leq V$ .

To show  $T$ -invariant, let  $x \in K_\lambda$ . Then, for period  $p \in \mathbb{Z}$  of  $x$ ,

$$(T - \lambda I)^p(T(x)) = ((T - \lambda I)^p \circ T)(x) = (T \circ (T - \lambda I)^p)(x) = T(0) = 0.$$

Thus  $T(x) \in K_\lambda$ . Including  $E_\lambda$  is clear.

Show 2). Let  $w \in K_\mu \cap K_\lambda$  be given. Let  $p, q \in \mathbb{Z}$  be period of  $w$  for  $\mu$  and  $\lambda$ , respectively.

Since  $\mu \neq \lambda$ ,  $(t - \mu)^p, (t - \lambda)^q \in F[t]$  are disjoint. Now, there exist  $g_1(t), g_2(t) \in F[t]$  s.t.

$$g_1(t)(t - \mu)^p + g_2(t)(t - \lambda)^q = 1.$$

because  $F[t]$  is P.I.D. Now,

$$(g_1(T)(T - \mu)^p + g_2(T)(T - \lambda)^q)(w) = 0 = w$$

Thus  $K_\mu \cap K_\lambda = \{0\}$ .

Finally, show 3). Since for any  $w \in K_\lambda$ ,  $(T - cI)(w) = 0 \Rightarrow w \in E_c \subseteq K_c$

By 2),  $w = 0$ . That is, kernel of  $(T - cI)|_{K_\lambda}$  is zero space, thus injective.

To show surjective, let  $x \in K_\lambda$ . For period  $p \in \mathbb{Z}$  of  $x$ ,

$$W = \{x, (T - \lambda I)(x), \dots, (T - \lambda I)^{p-1}(x)\} \subseteq K_\lambda$$

is  $(T - \lambda I)$ -invariant. Now,  $(T - \lambda I)|_W$  is injection on fin. dim space, thus surjection.

Now, we can find  $y \in W$  s.t.  $(T - \lambda I)(y) = x$ .

Thm. If  $f(t)$  splits over  $F$ , then for any eigenvalue  $\lambda$  with multiplicity  $m \in \mathbb{N}$ ,

$$\dim K_\lambda \leq m \text{ and } K_\lambda = \ker((T - \lambda I)^m)$$

proof. Let  $h(t)$  be a characteristic polynomial of a restriction  $T|_{K_\lambda}$ .

Since for any distinct eigenvalue  $\mu \neq \lambda$ ,  $K_\mu \cap K_\lambda = \{0\}$ ,

$\lambda$  is a unique eigenvalue of  $T|_{K_\lambda}$ . Moreover, since  $h(t)$  divides  $f(t)$ ,

$$h(t) = (\lambda - t)^d \quad (d \leq m).$$

Clearly,  $\ker(T - \lambda I)^m \subseteq K_\lambda$ .

Meanwhile, by assumption,  $f(t) = (t - \lambda)^m \cdot g(t)$  where  $g(\lambda) \neq 0$ .

Therefore,  $g(T)|_{K_\lambda}$  is bijective. Now, for any  $x \in K_\lambda$ , there is  $y \in K_\lambda$  s.t.  $g(T)(y) = x$ .

The Cayley-Hamilton Thm gives

$$(T - \lambda I)^m(x) = (T - \lambda I)^m(g(T)(y)) = f(T)(y) = T_0(y) = 0.$$

Thus  $x \in \ker(T - \lambda I)^m$ .

$$\text{Im } U = (T - \lambda I)|_{K_\lambda} [K_\lambda]$$

$$= (T - \lambda I)|_{K_\lambda} \left[ \bigcup_{n=1}^{\infty} \ker(T - \lambda I)^n \right]$$

$$= \bigcup_{n=1}^{\infty} (T - \lambda I) [\ker(T - \lambda I)^n]$$

$$\begin{array}{c} V \\ | \\ K_\lambda \\ | \\ \text{Im } U \end{array}$$

$$T|_{\text{Im } U}$$